Burrowing Into Extinction

Prey in our simulation evolve a claw that lets them hide underground. Every burrow drains the soil that plants need to grow. Once the claw takes over, the prey run out of food and die out.

Summary How burrowing works Does it count? Main experiment Where it happens How we got here Settings Limitations Reproducing

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21 of 40
runs where claws took over and then every prey died, with the current settings

0 of 20
with the same settings when burrowing doesn't damage the soil (0 of 124 harmless runs across all sweeps)

17 of 20
runs where prey survive 100,000 ticks when the claw can't burrow

p = 0.00002
Fisher's exact test, 21 of 40 against 0 of 20

Summary

Evolutionary suicide is when natural selection favours a trait that ends up driving its own carriers extinct. We built a version of it into LifeEngine. A **claw cell** lets prey burrow when they see a predator. Burrowing protects them, but each burrow removes fertility from the soil underneath, and plants can only grow on fertile soil.

No organism starts with a claw. Claws appear by mutation, and with the current settings they take over the prey population in most runs. When burrowing damages the soil, the soil falls to about 70% fertility, the plants collapse and every prey dies in about half the runs. When the same burrowing is harmless, claws spread just as well and nothing collapses. Without working claws, prey survive in 17 of 20 runs, so the collapse is caused by the trait and not by a world that was failing anyway.

The effect only appears in a window of settings. Predators must be strong enough to make burrowing pay off but not so strong that they wipe out the prey on their own. Burrows must also be damaging enough to wreck the soil but not so damaging that the claw is useless to its owner. Within that window the result is stable: moderate changes to hide duration, stay hidden range and soil regrowth don't change it meaningfully.

All results come from 456 runs of 100,000 ticks each, with 20 runs per setup in the main experiments. They were run headless in Node, using the same `src/` code as the browser.

How burrowing works

- **Getting a claw.** No prey start with one. A claw appears when a mutation adds or changes a cell; about 1 in 7 new cells is a claw. It costs food to reproduce like any other cell.
- **Burrowing.** A prey with a claw that sees a predator burrows instead of running, as long as the soil under it has an average fertility of at least 1.
- **While hidden.** It is invisible to predators for 100 ticks and doesn't age. If a predator is still within 6 cells when the time is up, it stays down for up to another 100 ticks.
- **The cost.** Each burrow removes 10 fertility from every soil cell under the prey. Soil holds at most 10, so one burrow empties that patch.
- **Recovery.** Each soil cell not covered by an organism has a 0.05% chance per tick to regain 1 fertility, so a drained patch takes thousands of ticks to recover.
- **The link to food.** Plants are only born where the soil under them averages at least 7. Prey eat the food plants produce.
- **Predators.** Any prey that grows a killer cell becomes a predator, claw or not. Predators live twice as long as other organisms of the same size.

Does it count as evolutionary suicide?

Each part of the usual definition, with the evidence for it at the current settings.

The definition requires	What the simulation shows	Met?
The population survives without the trait	With burrowing disabled, prey survive 100k ticks in 17 of 20 runs	**Yes**
The trait arises by mutation and is favoured	No seeded claws, and claws cost food. Claws still take over the prey population in 19 of 20 runs when burrowing is harmless	**Yes**
The trait degrades a shared resource	With damaging burrowing the soil falls to a median low of 70%, and plants fall to a median low of 23 against 408 with harmless burrowing	**Yes**
The degradation drives the carriers extinct	All prey die after take-over in 21 of 40 damaging runs and 0 of 20 harmless ones ($p = 0.00002$)	**Yes**
Selection keeps favouring the trait as it fails	Not measured directly. In collapsing runs the claw has already replaced the clawless prey, so no alternative remains	**Untested**

Main experiment

Five setups, each run from the same 20 starting layouts (placement seeds 1 to 20) for 100,000 ticks: burrowing disabled, harmless burrowing, and burrowing that costs the soil 3, 5 or 10 per burrow. Everything else is at the current defaults. Every run is sorted into one outcome; claws count as having *taken over* once there are at least 50 prey with a claw and they make up at least half of all prey.

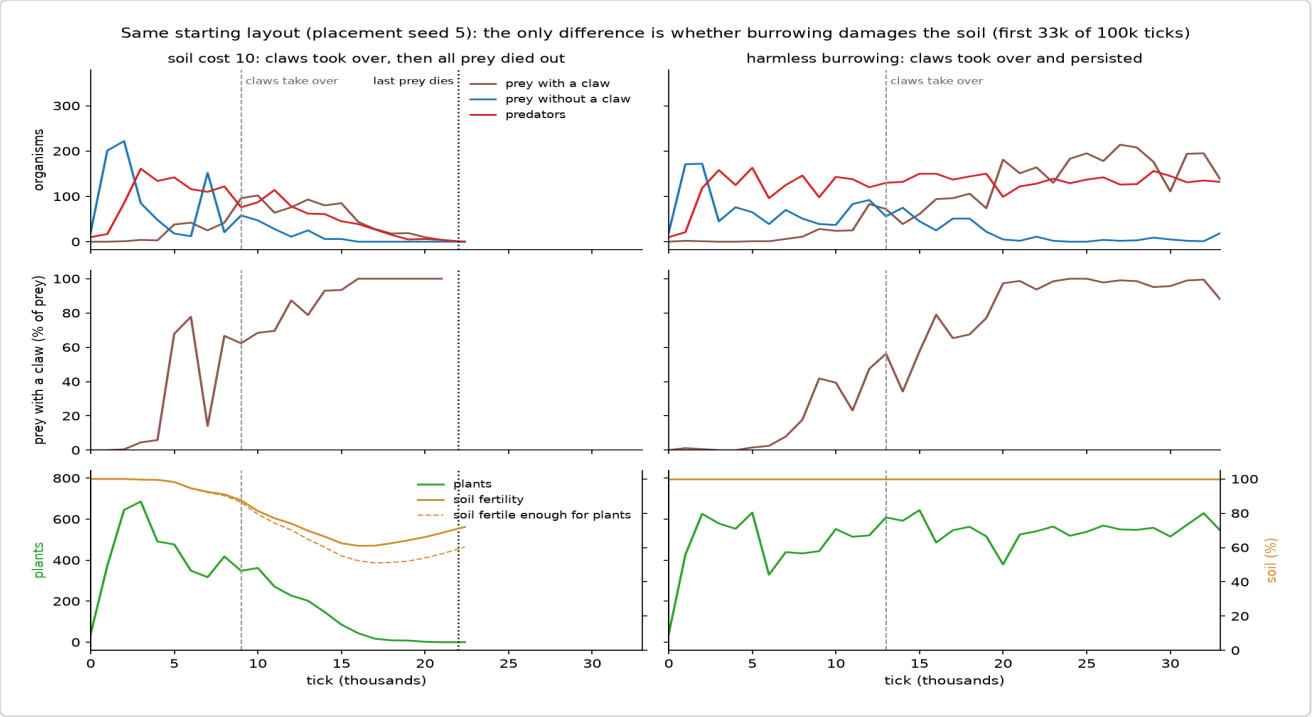


Figure 1. One starting layout run with damaging and with harmless burrowing. On the left, claws take over at 9,000 ticks, the soil drains to about 60% and the plants collapse. Every prey is dead by 22,000 ticks and every organism by 22,400. On the right, claws take over at 13,000 ticks and stay; the soil never changes and plants hold at about 550.

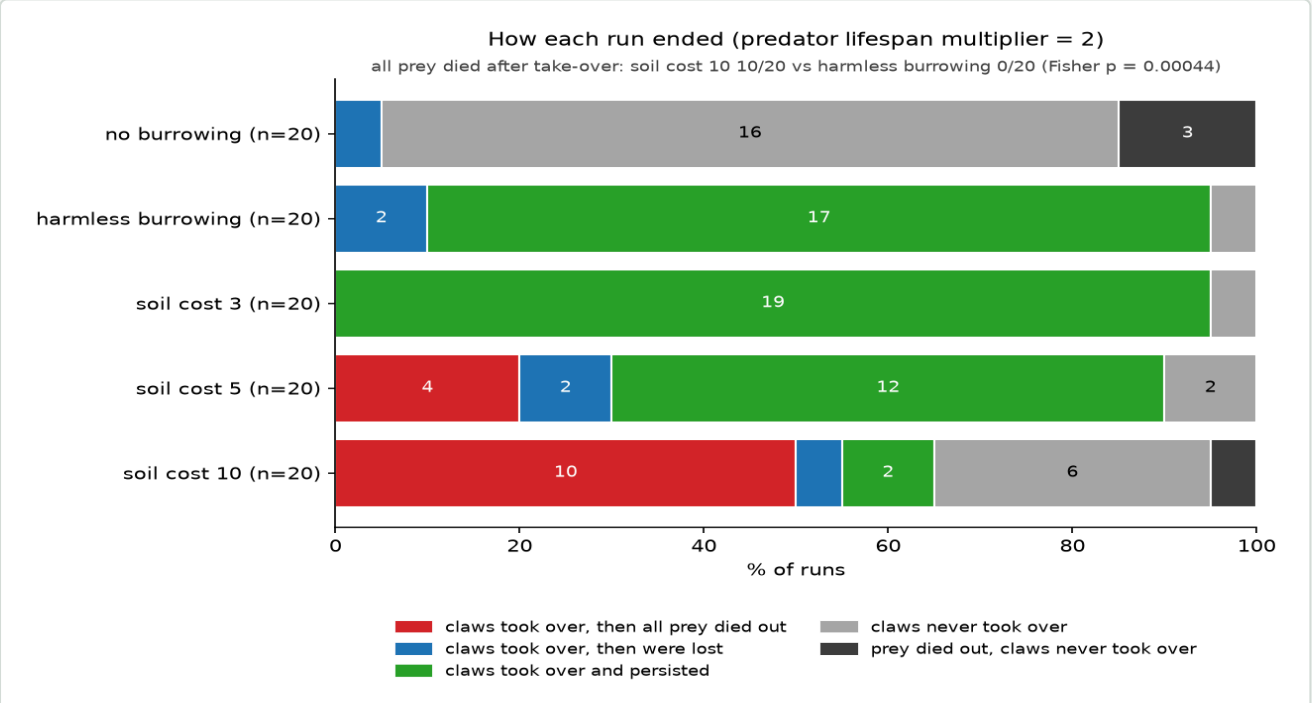


Figure 2. How each run ended. Harmless burrowing and soil cost 3 never collapse; in almost every run the claws take over and stay. Soil cost 5 collapses in 4 of 20 runs. Soil cost 10 collapses in 10 of 20, although its claws take over less often (13 of 20), probably because each burrower empties its own patch in one go.

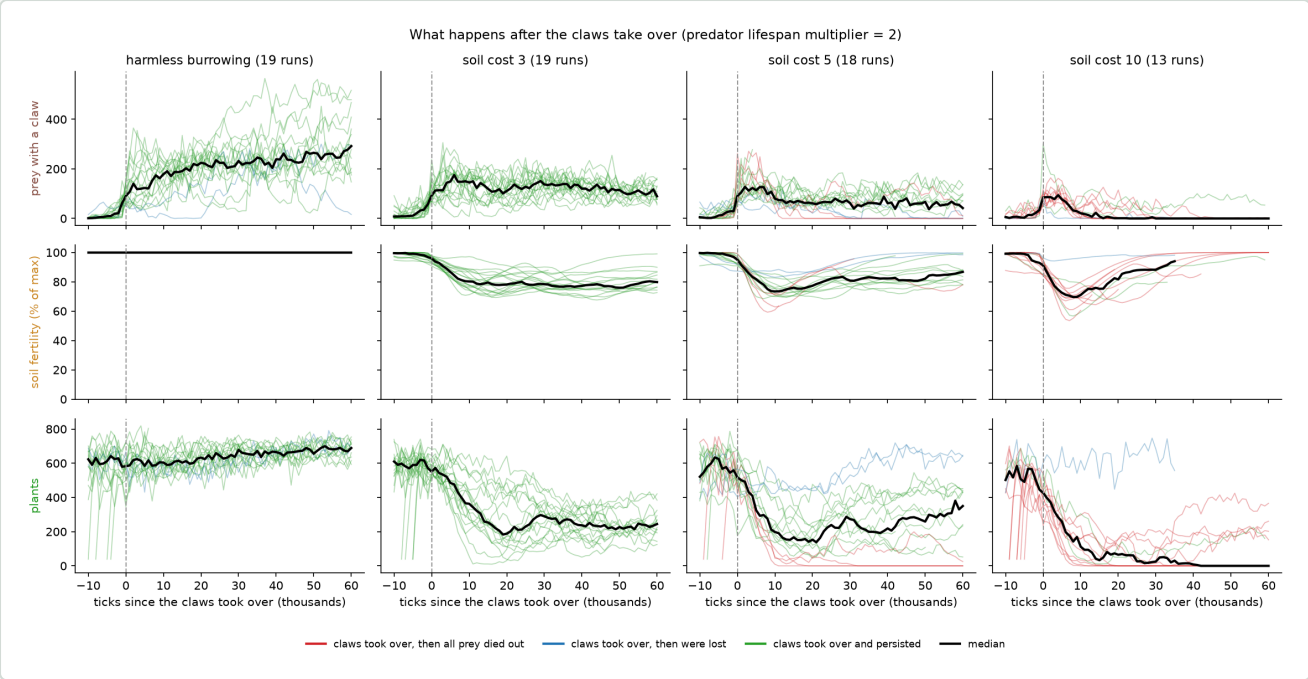


Figure 3. Every run lined up on the moment claws took over (dashed line), with the median in black. This is the clearest comparison: with harmless burrowing the plants never notice the claws. With soil cost 3 the soil settles near 78% and the plants at about 250, a damaged but stable world. With soil cost 10 the plants fall to zero within about 40,000 ticks of the take-over, and the claw prey go with them.

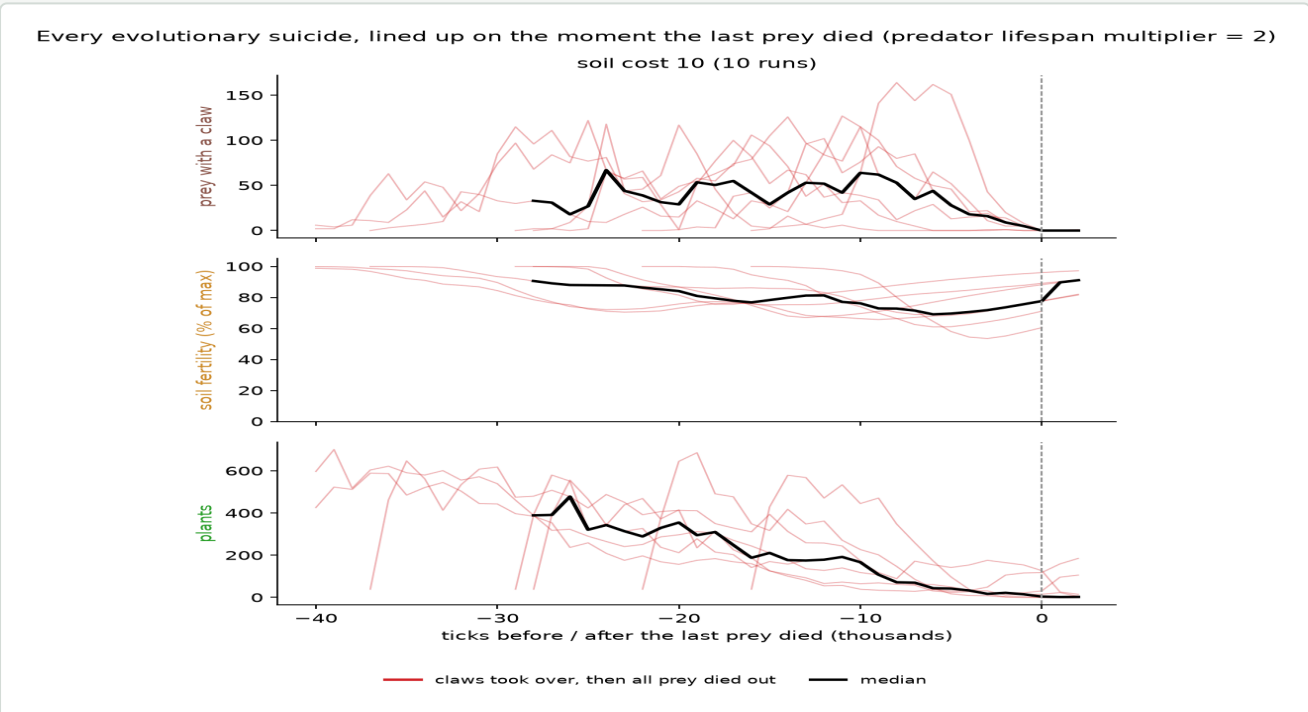


Figure 4. The 10 collapses at soil cost 10, lined up on the moment the last prey died. They follow the same order: the soil drains while claw prey are common, the plants decline, then the prey die out. The soil starts recovering near the end as burrowers become rare, but the plants have already gone.

How the prey die. In the runs where we recorded causes of death, predators caused 70 to 97% of claw-prey deaths, in harmless runs as well as collapsing ones, so the cause of each death doesn't separate the setups. What separates them is whether the plants recover. Once they don't, prey can't collect the food they need to reproduce, and predation and old age finish them off.

Where it happens

Three settings decide whether the claw spreads and whether its spread is fatal. In the tables, each cell shows how many runs the claws took over in, then how many ended with every prey dead.

Predator pressure

The predator lifespan multiplier sets how long predators live. At 3, predators kill every prey in 7 of 8 runs even when the claw can't burrow, so a collapse there can't be blamed on the claw. At 1.5 and 1, predators are weaker, prey burrow less and collapses are rare (2 and 1 of 8 runs). At 2, the world without working claws survives and the damaging claw still collapses it.

Predator lifespan ×	Prey survive, no burrowing	Harmless	Soil cost 3	Soil cost 10
3	1 / 8	7 / 8 · 0	1 / 8 · 0	5 / 8 · 5
2 (current)	17 / 20	19 / 20 · 0	19 / 20 · 0	13 / 20 · 10
1.5	8 / 8	6 / 8 · 0	7 / 8 · 0	7 / 8 · 2
1	8 / 8	4 / 8 · 0	5 / 8 · 0	4 / 8 · 1

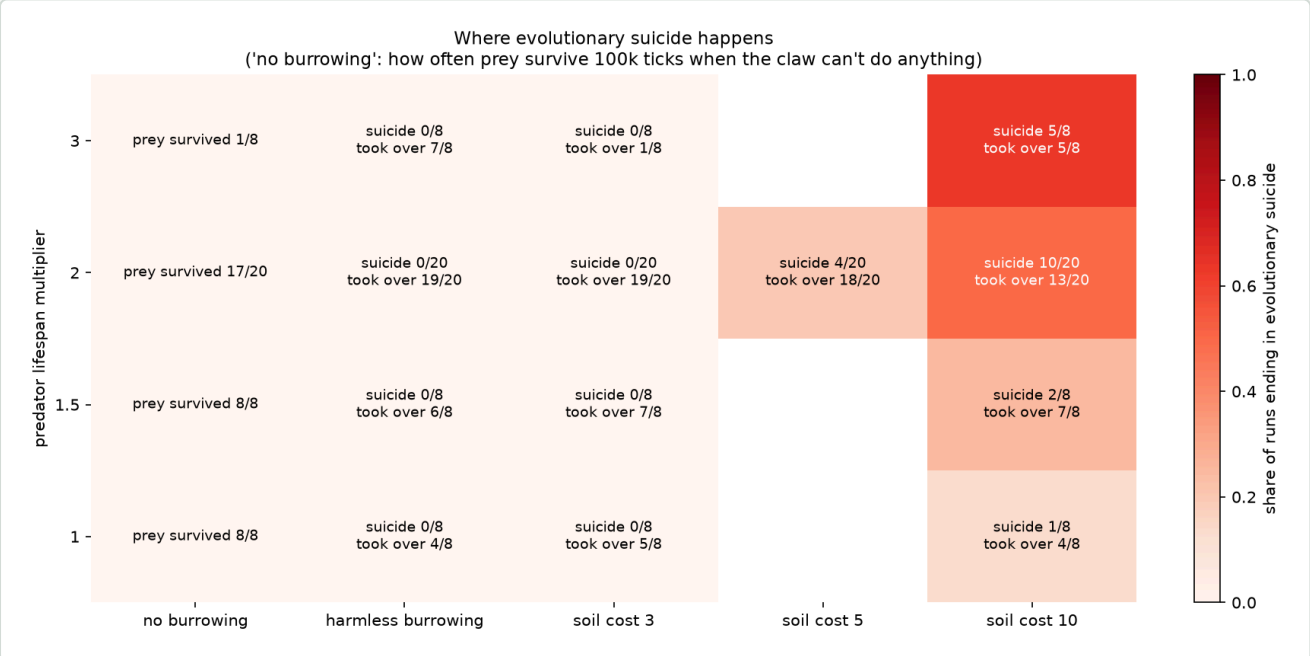


Figure 5. Share of runs that end in evolutionary suicide, by predator lifespan and setup. The multiplier-2 row uses 20 runs per cell; the others use 8.

Soil cost per burrow

The soil cost trades off two things. A cheap burrow lets prey burrow on the same patch repeatedly, so the claw spreads easily, but the damage is slow enough for the soil to settle at about 75 to 80% and the population carries on.

An expensive burrow wrecks the soil faster but empties the burrower's own patch at once, so the claw pays off less and takes over less often. At the current hide duration, collapses rise from 0 of 20 runs at cost 3, to 4 of 20 at cost 5 ($p = 0.03$ against cost 10), to 21 of 40 at cost 10. Cost 7 gave 6 of 20 with slower collapses (median 83,000 ticks), but that isn't clearly different from cost 10 ($p = 0.17$).

Hide duration

We tested 30-tick hides with soil cost 5, a combination suggested from browser runs. With 30-tick hides, each claw prey burrows about twice as often (6 to 8 times per 1,000 ticks against about 4), so an individual burrower damages the soil faster. But it spends only 23 to 32% of its time underground instead of 40 to 51%, which means less protection and less time without ageing. The claw no longer earns back its food cost and rarely spreads: with harmless burrowing, an average of 2 claw prey are alive at a time, against 157 with 100-tick hides. A trait that stays rare can't wreck the soil.

Hide duration	Harmless	Soil cost 3	Soil cost 5	Soil cost 10
100 (current)	19 / 20 · 0	19 / 20 · 0	18 / 20 · 4	13 / 20 · 10
30	2 / 20 · 0	3 / 20 · 0	7 / 20 · 2	3 / 20 · 2

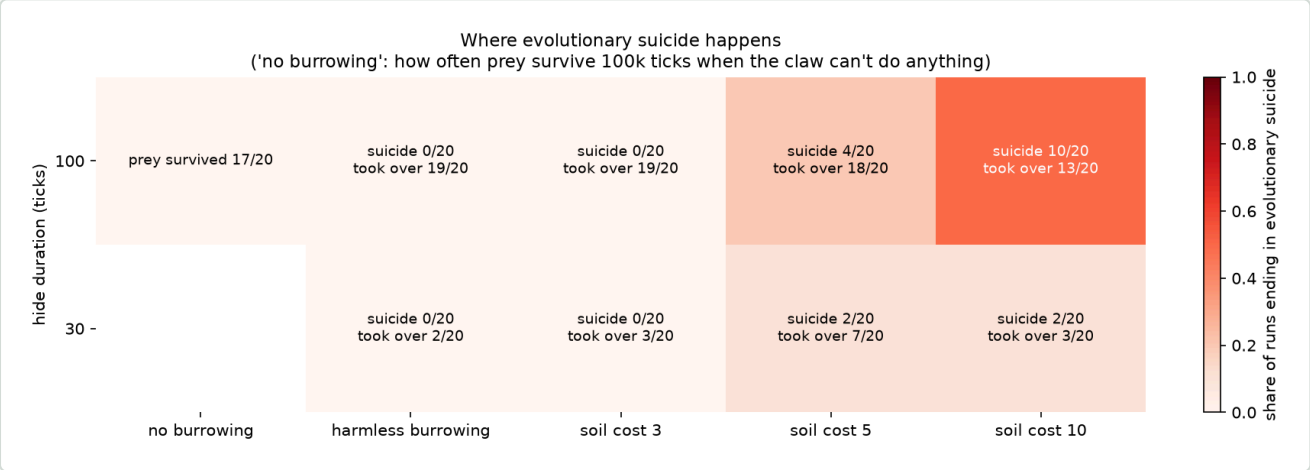


Figure 6. Hide duration 100 against 30, 20 runs per cell. Short hides stop the claw from spreading, so there is little to collapse.

Fine-tuning

Starting from the current settings, we changed one thing at a time, 20 runs each. The changes that alter what the claw is worth to a prey also got a harmless-burrowing version. The current settings were rerun in the same batch, which gave 11 of 20, matching the first batch's 10 of 20.

Setup	All prey died	Claws took over	Median collapse tick	p against current (21 / 40)
Current settings (this batch)	11 / 20	16 / 20	44,000	—
Hide duration 150	10 / 20	18 / 20	34,000	1.0
Soil cost 7	6 / 20	17 / 20	83,000	0.17
Soil regrowth 0.025 (half as fast)	13 / 20	17 / 20	46,000	0.42
Stay hidden range 3	11 / 20	18 / 20	59,000	1.0
Stay hidden range 10	12 / 20	19 / 20	52,000	0.78
Harmless, with hide 150, range 3 or range 10	0 / 20 each	18 to 19 / 20	—	—

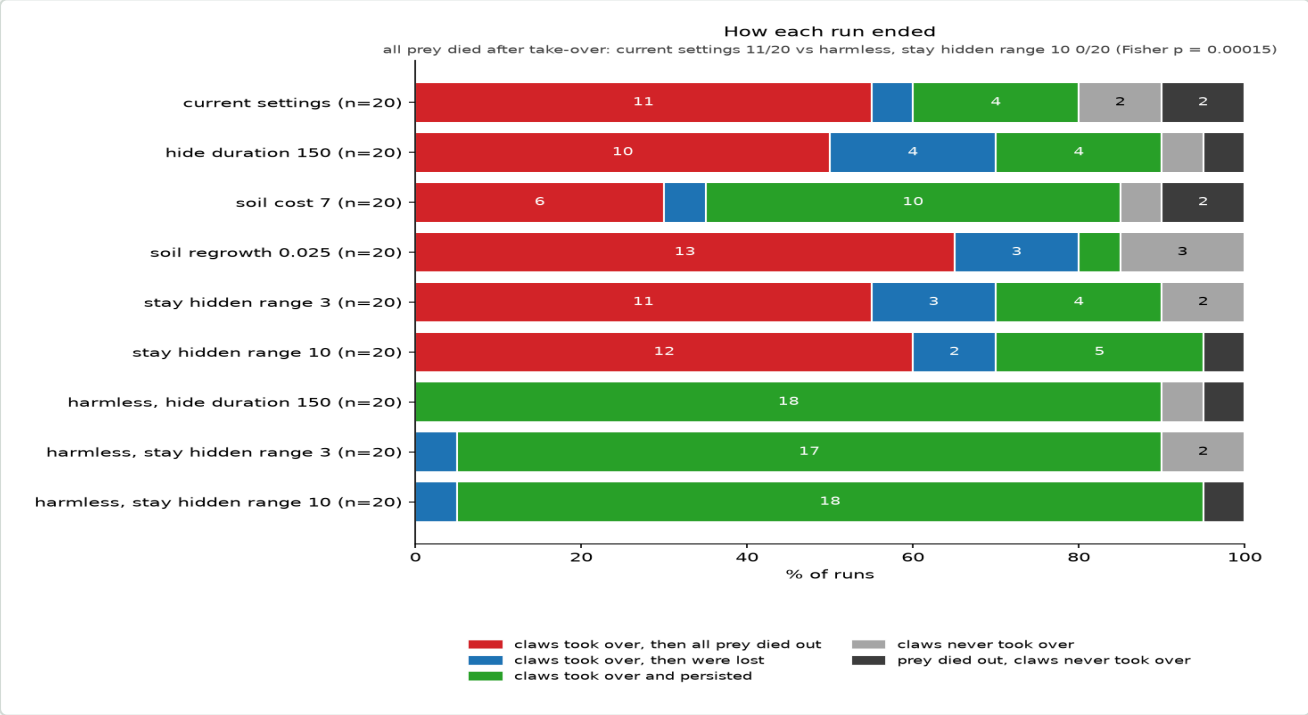


Figure 7. How each tuning run ended. None of the changes clearly beats the current settings. Slower soil regrowth scored highest, and of its 17 take-overs only 1 persisted. With 20 runs per setup, though, that could be chance.

How we got here

These steps used earlier versions of the code and an exploratory test harness with 6 to 12 runs per setup, so the numbers are rougher than in the rest of the report.

1. **The first burrowers died before damaging anything.** The claw cost food to reproduce, burrowers moved slowly, they hid once and resurfaced next to the predator, and many started on soil too poor to burrow in. The soil never degraded.
2. **We made the claw worth having.** We added free claw cells, no ageing while hidden, staying hidden while a predator is close, and burrowing only when a predator is near, and made burrowers move every tick. With 10 claw prey seeded at the start, burrowers died out in 6 of 8 runs by 60,000 ticks and never with harmless burrowing. The claw was seeded rather than evolved, though, and predators wiped out regular prey early.
3. **Teammates' changes made the controls cleaner.** Predators can now re-evolve from prey, starting organisms are placed at seeded random spots, and every moving organism that isn't a predator counts as prey. After these, harmless burrowing never collapsed (0 of 8 runs) and plants never fell below 276.
4. **Claws had to evolve on their own.** We removed the seeded claws and made claw cells cost food again, because free claws also helped plants and predators. Claws appeared by mutation within 2,000 ticks in every run. But at predator lifespan 3 the predator boom killed all prey in about half the runs before claws mattered, and with soil damage on, claws never spread far.
5. **Turning off "Only burrow when predator is close"** produced the first clean collapse from evolved claws in 40,000-tick runs. The world without claws still failed too often.
6. **A 100,000-tick sweep over predator lifespan** found the window. At 2, prey survive without the claw while the damaging claw still collapses the world.
7. **The headless scripts** confirmed this with 20 runs per setup and tested hide duration, soil cost, soil regrowth and stay hidden range.

Recommended settings

These are the current defaults on the `tarique-testing` branch.

Setting	Value	Why	Evidence
Starting organisms	`10 predators, 20 prey, 0 claw prey, 39 plants`	The claw must evolve rather than be seeded	Design choice
Claw cells are free	`off`	Free claws spread for reasons unrelated to burrowing	Design choice
Clawed killer becomes predator	`1`	One rule for every prey; little effect on results	16 runs, 40k ticks
Predator lifespan multiplier	`2`	The only value where both the baseline survives and collapses happen	128 runs
Only burrow when predator is close	`off`	Keeps the soil damage going	8 runs per setting, 40k ticks
Hide fertility cost	`10`	Most collapses	20 to 40 runs per value
Hide duration	`100`	At 30 the claw barely spreads; 150 changes nothing	20 runs per value
Don't age while hidden, stay hidden near predators	`on, range 6`	Part of what makes the claw worth having; range 3 or 10 changes nothing	20 runs per range; never switched off
Soil regrowth chance	`0.05%`	0.025% looked slightly better but not convincingly	20 runs
Min burrow fertility, burrower move interval, predator move interval, min plant fertility, lifespan multiplier	`1, 1, 3, 7, 150`	Left at defaults	Not varied

Limitations

- **Headless runs.** The scripts drive the same `src/` code as the browser, but we haven't checked these rates against browser runs.
- **One soil layout.** Every run used soil seed 42 on a 288 by 180 grid; only the placement of the starting organisms varied.
- **Sample size.** With 20 runs per setup, a difference of 2 or 3 collapses could be chance. Only the large effects in this report are statistically clear.
- **Proximate cause of death.** Predators kill most prey in every setup. The soil acts indirectly, by stopping the plants from recovering.

- **Collapse takes everything.** Predators usually die out after the prey, and plants often do too. In some runs new prey later evolve from plants; we count the first time every prey is dead.
- **Outcome rules.** A take-over needs at least 50 claw prey making up at least half of all prey. A run that ends with 2 or fewer prey and 5 or fewer plants counts as collapsed, since it can't recover.
- **Missing measurement.** We didn't compare the fitness of clawed and clawless prey during the decline, so we can't yet show that selection still favours the claw as the population fails.

Reproducing the results

Everything is in `analysis/headless/`. A 100,000-tick run takes about 1.5 minutes, and the sweep script runs one per CPU core, so 80 runs take about 30 minutes on an 8-core laptop. An interrupted sweep picks up where it stopped. Plotting needs `pip install matplotlib numpy`.

```
 # run the sweeps (results go to analysis/headless/results/<sweep name>/)
node analysis/headless/sweep.js analysis/headless/sweeps/evo_suicide.json
node analysis/headless/sweep.js analysis/headless/sweeps/predator_sweep.json
node analysis/headless/sweep.js analysis/headless/sweeps/hide_sweep.json
node analysis/headless/sweep.js analysis/headless/sweeps/tuning_sweep.json

# draw the figures
python analysis/headless/plot_evo_suicide.py analysis/headless/results/evo_suicide
analysis/headless/results/predator_sweep
python analysis/headless/plot_evo_suicide.py analysis/headless/results/hide_sweep
analysis/headless/results/evo_suicide \
    --by hideDuration --focus 30 --damage "soil cost 5"
python analysis/headless/plot_evo_suicide.py analysis/headless/results/tuning_sweep \
    --damage "current settings" --control "harmless, stay hidden range 10"

# a single run with any setting from Run Setup or Evolution Controls
node analysis/headless/run_sim.js --ticks 100000 --set '{"hideFertilityCost":0,"placementSeed":3}' --out
run.json
```

Runs: 100 in `evo_suicide`, 96 in `predator_sweep`, 80 in `hide_sweep`, 180 in `tuning_sweep`, all 100,000 ticks. Significance tests are two-sided Fisher's exact tests on the number of runs where every prey died after the claws took over.